Part 2 – Analytics, Monitoring & Growth Intelligence

7. Performance Analytics, Monitoring & Leaderboard Management

This module provides comprehensive class-level and student-level performance tracking, trend analysis, growth monitoring, and ranking mechanisms. All calculations and formulas must remain consistent with the Student Portal logic to ensure data integrity across the platform.

7.1 Class Performance Analytics

Teachers must be able to view real-time class-level metrics including:

- Average Quiz Score
- Average Assignment Score
- Submission Consistency
- Module Completion Rate
- Overall Class OGI
- Module-wise Attendance Comparison
- Weekly Activity Trend
- Attendance Heatmap

These analytics must aggregate data across batches and courses where applicable.

7.2 Student Performance Monitoring

Teachers must be able to monitor individual student performance including:

- OGI trend over time
- Growth classification (Excellent, Improving, Stable, Needs Attention)
- Module completion progress
- Weekly learning hours
- Skill acquisition count
- Learning streak

Teachers must be able to open detailed student insights directly from analytics dashboards.

Optional:

- Download individual student performance report (PDF or CSV format)

7.3 Weekly Performance Snapshot System

The system must automatically generate weekly aggregated data per student, including:

- Quiz average
- Assignment average
- Completion rate
- Submission consistency
- Overall Growth Index (OGI)

Teachers must be able to:

- Compare week-to-week performance
- Identify improvement trends
- Identify declining performance
- Detect performance stagnation

Snapshots must be stored historically for trend visualization.

7.4 Leaderboard Management

Teachers must be able to:

- View class leaderboard
- Filter by Batch

- Filter by Course
- Sort by:
    - OGI
    - Attendance
    - Assignment Score

The leaderboard must be calculated using:

- Overall OGI

  or
- A weighted performance score (if configured)

Leaderboard data must update automatically whenever:

- Assignment evaluations occur
- Quiz results are published
- Attendance records change
- Module completion updates

Real-Time Synchronization Requirement

The following must trigger real-time updates:

- Assignment evaluation
- Quiz result publication
- Attendance update
- Module completion
- Student enrollment changes

All dashboards and analytics must reflect the latest data instantly.